Hedonic Rent Regressions: Raw Long-Run Filing Rate

Philly Evictions Project

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1 Overview

This document summarizes the hedonic rent regressions from `r/price-regs.R`. The script estimates the relationship between eviction filing intensity (raw long-run filing rate, pre-2019) and rent levels, controlling for building and neighborhood characteristics.

Unit of analysis: Building (PID) × year panel, 2014–2019.

Key data inputs:

- `analytic_sample.csv` — PID × year panel with rents, filing rates, building characteristics, and tenant demographics

Sample restrictions:

- Years 2014–2019 (pre-COVID hedonic)
- Non-missing raw filing rate (pre-2019)
- Filing rate ≤ 0.75 (excludes extreme outliers)

Key measure: `filing_rate_longrun_pre2019` = total pre-2019 filings / total units / years observed pre-2019. This is the unshrunk filing rate (contrast with the Empirical Bayes version in `price-regs-eb.R`).

2 Eviction Intensity Measures

Two representations of eviction filing intensity are used:

- **Binned:** Pre-2019 raw filing rate cut at 0, 5%, 10%, 20%, with reference category (5–10%]
- **Continuous:** Raw filing rate (pre-2019), entering linearly

3 Empirical Strategy

3.1 Fixed Effects and Controls

All specifications include a rich set of fixed effects:

- **Year** (absorbs rent trends)
- **GEOID** (census tract — absorbs neighborhood-level rent premia)
- **Year built decade** (construction vintage)
- **Rent source** (Altos vs other)
- **Unit count bin** (building size category)
- **Building type** (structural type)
- **Quality grade** (A/B/C/D, standardized)
- **Number of stories bin**

Additional controls: log(total area), log(market value).

Regressions are weighted by total units and clustered by PID.

3.2 Model 1: Binned Filing Rate (Baseline)

$$\log(\text{rent}_{it}) = \sum_b \beta_b \cdot \mathbb{1}[\text{rate bin}_i = b] + \gamma_1 \cdot \text{ViolFlag}_i + \gamma_2 \cdot \text{ComplaintFlag}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

where ViolFlag_i indicates any unsafe/dangerous/hazardous violation and ComplaintFlag_i indicates any heat/fire/drainage/plumbing complaint. Reference bin: (5–10%].

3.3 Model 2: Continuous Filing Rate

$$\log(\text{rent}_{it}) = \beta \cdot \text{rate}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

3.4 Models 3–4: Tenant Composition Controls

Models 1 and 2 augmented with additive tenant composition controls:

- `infousa_pct_black_imp` (share Black, mean-imputed)
- `infousa_pct_female_imp` (share female HoH, mean-imputed)
- `infousa_pct_black_female_imp` (share Black female HoH)
- `infousa_share_persons_demog_ok_imp` (demographic coverage)
- `tenant_comp_missing` (indicator for missing composition)

3.5 Model 5: Complaint × High-Eviction Interaction

$$\log(\text{rent}_{it}) = \beta_1 \cdot \text{ComplaintFlag}_i + \beta_2 \cdot \text{HighEvict}_i + \beta_3 \cdot \text{ComplaintFlag}_i \times \text{HighEvict}_i + \delta' X_i + \alpha_{\text{tract}} + \phi_t + \varepsilon_{it}$$

where $\text{HighEvict}_i = \mathbb{1}[\text{rate}_i > 0.20]$. Uses a sparser FE set (year + GEOID + unit count bin).

4 Results

4.1 Baseline: Binned and Continuous Specifications

Dependent Variable:	log_med_rent	
Model:	(1)	(2)
Variables		
Filing rate = 0	0.0953***	
	(0.0118)	
Filing rate (0, 5%]	0.0477***	
	(0.0130)	
Filing rate (10, 20%]	-0.0295***	
	(0.0110)	
Filing rate 20%+	-0.0377***	
	(0.0097)	
Fixed-effects		
year	Yes	Yes
GEOID	Yes	Yes
year_blt_decade	Yes	Yes
source	Yes	Yes
num_units_bin	Yes	Yes
building_type	Yes	Yes
quality_grade_standard	Yes	Yes
num_stories_bin	Yes	Yes
Fit statistics		
Observations	116,641	116,641
R^2	0.67288	0.66984

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

4.2 Full Table: With Tenant Composition Controls

Dependent Variable:	log_med_rent			
Model:	(1)	(2)	(3)	(4)
Variables				
Filing rate = 0	0.0953***		0.0911***	
	(0.0118)		(0.0117)	
Filing rate (0, 5%]	0.0477***		0.0463***	
	(0.0130)		(0.0129)	
Filing rate (10, 20%]	-0.0295***		-0.0261**	
	(0.0110)		(0.0110)	
Filing rate 20%+	-0.0377***		-0.0343***	
	(0.0097)		(0.0096)	
any_unsafe_dangerous_violationTRUE	-0.0076		-0.0060	
	(0.0156)		(0.0155)	
any_heat_fire_drainage_plumbing_complaintTRUE	-0.0263*		-0.0244	
	(0.0150)		(0.0149)	
log(total_area)	0.0079	0.0079	0.0080	0.0080
	(0.0075)	(0.0077)	(0.0075)	(0.0076)
log(market_value)	0.0533***	0.0523***	0.0528***	0.0519***
	(0.0091)	(0.0090)	(0.0091)	(0.0090)
Filing rate (pre-2019)		-0.3270***		-0.3024***
		(0.0284)		(0.0280)
infousa_pct_black_imp			-0.1143***	-0.1159***
			(0.0156)	(0.0157)
infousa_pct_female_imp			-0.0592***	-0.0610***
			(0.0101)	(0.0103)
infousa_pct_black_female_imp			0.0574***	0.0585***
			(0.0135)	(0.0136)
infousa_share_persons_demog_ok_imp			0.0327**	0.0398***
			(0.0152)	(0.0154)
tenant_comp_missing			0.0382***	0.0454***
			(0.0091)	(0.0091)
Fixed-effects				
year	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
source	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
building_type	Yes	Yes	Yes	Yes
quality_grade_standard	Yes	Yes	Yes	Yes
num_stories_bin	Yes	Yes	Yes	Yes
Fit statistics				
Observations	116,641	116,641	116,641	116,641
R^2	0.67288	0.66984	0.67480	0.67209

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

4.3 Coefficient Comparison: Filing Rate Across Specifications

Table 1: Filing rate coefficients: baseline vs tenant-composition controlled

Model	Term	Estimate	SE	p
baseline_bin	filing_rate_bin0	0.0953	0.0118	0.000
baseline_bin	filing_rate_bin(0-5%]	0.0477	0.0130	0.000
baseline_bin	filing_rate_bin(10-20%]	-0.0295	0.0110	0.008
baseline_bin	filing_rate_bin20%+	-0.0377	0.0097	0.000
tenant_bin	filing_rate_bin0	0.0911	0.0117	0.000
tenant_bin	filing_rate_bin(0-5%]	0.0463	0.0129	0.000
tenant_bin	filing_rate_bin(10-20%]	-0.0261	0.0110	0.018
tenant_bin	filing_rate_bin20%+	-0.0343	0.0096	0.000
baseline_cont	filing_rate_longrun_pre2019	-0.3270	0.0284	0.000
tenant_cont	filing_rate_longrun_pre2019	-0.3024	0.0280	0.000

4.4 Coefficient Plot: Binned Model

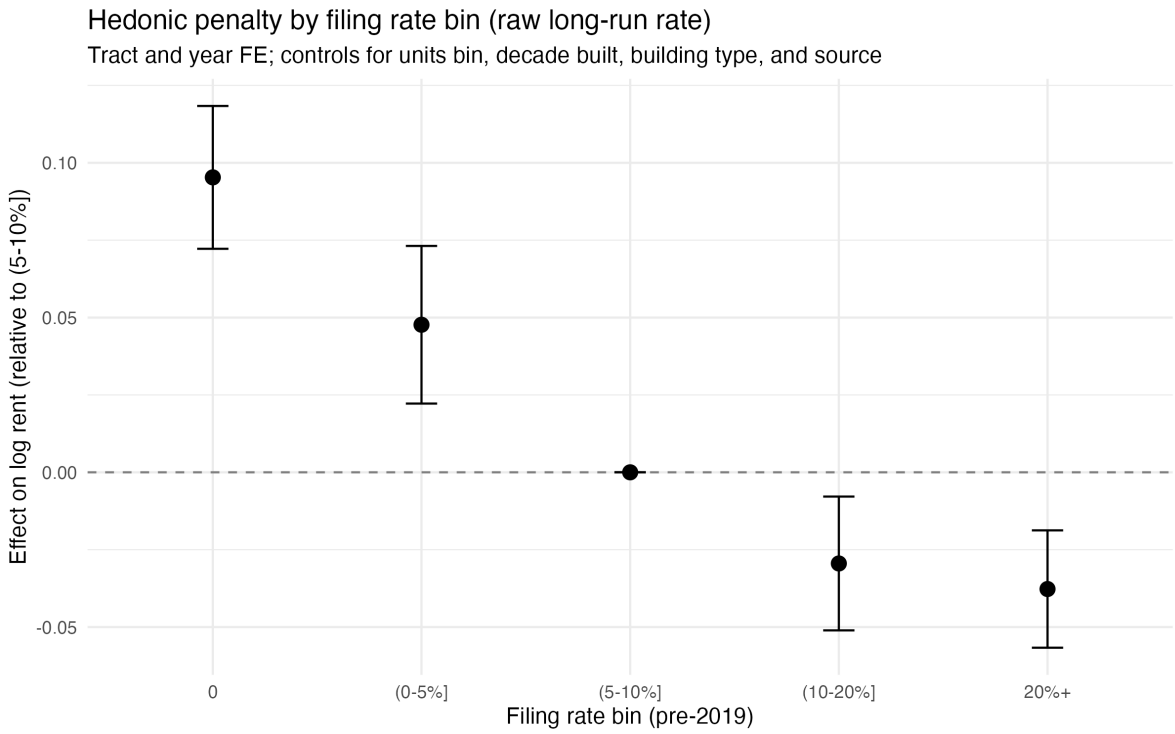

Figure 1: Hedonic rent penalty by raw filing rate bin (baseline specification). Reference category: 5-10 percent filing rate bin. Tract and year FE with building controls.

5 Exploratory Diagnostics

These plots residualize both rent and filing rate on the same controls and fixed effects used in the hedonic regressions, then examine the remaining relationship.

5.1 Residual Variation in Raw Filing Rate

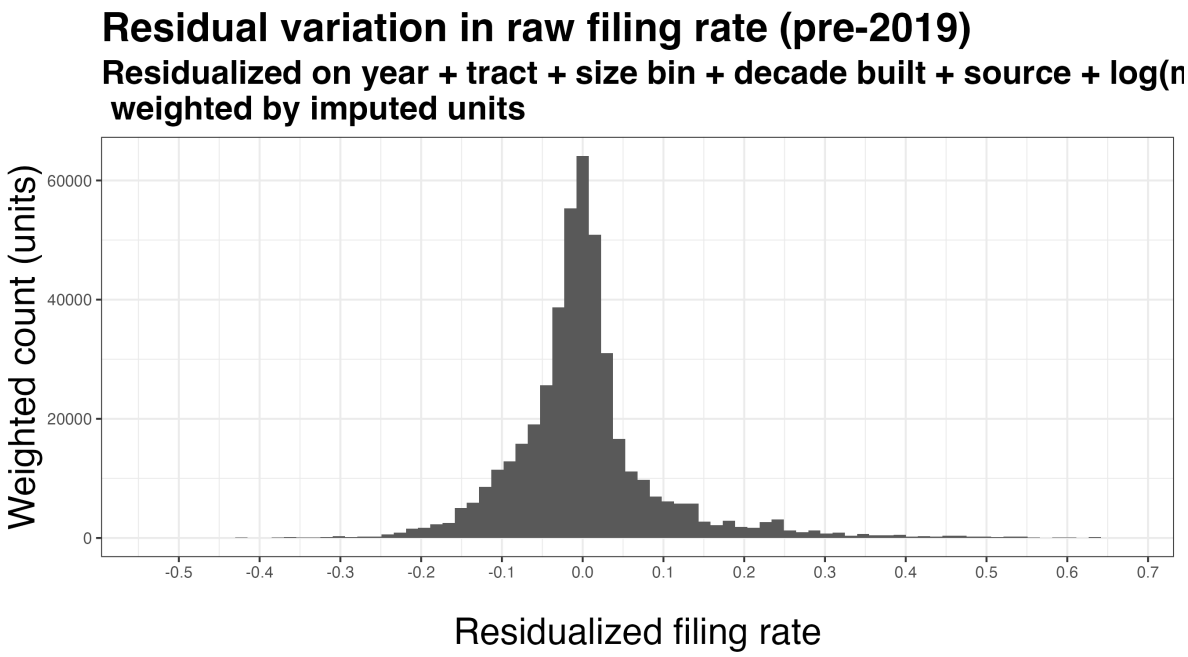

Figure 2: Distribution of residualized raw filing rate (pre-2019). Residualized on year + tract + building controls; weighted by total units.

5.2 Residualized Rent vs Filing Rate: Bin Scatter

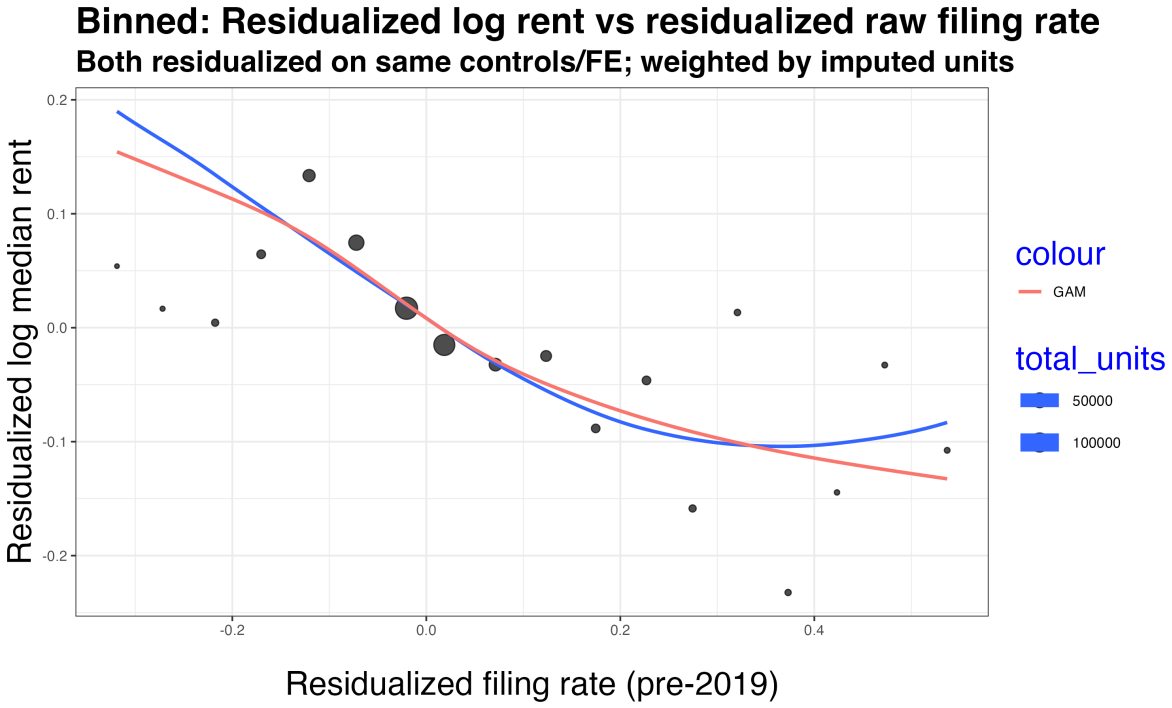

Figure 3: Binned scatter of residualized log rent vs residualized raw filing rate. Both residualized on the same controls/FE; weighted by total units.

5.3 Spline-Implied Stigma Curve (Cubic)

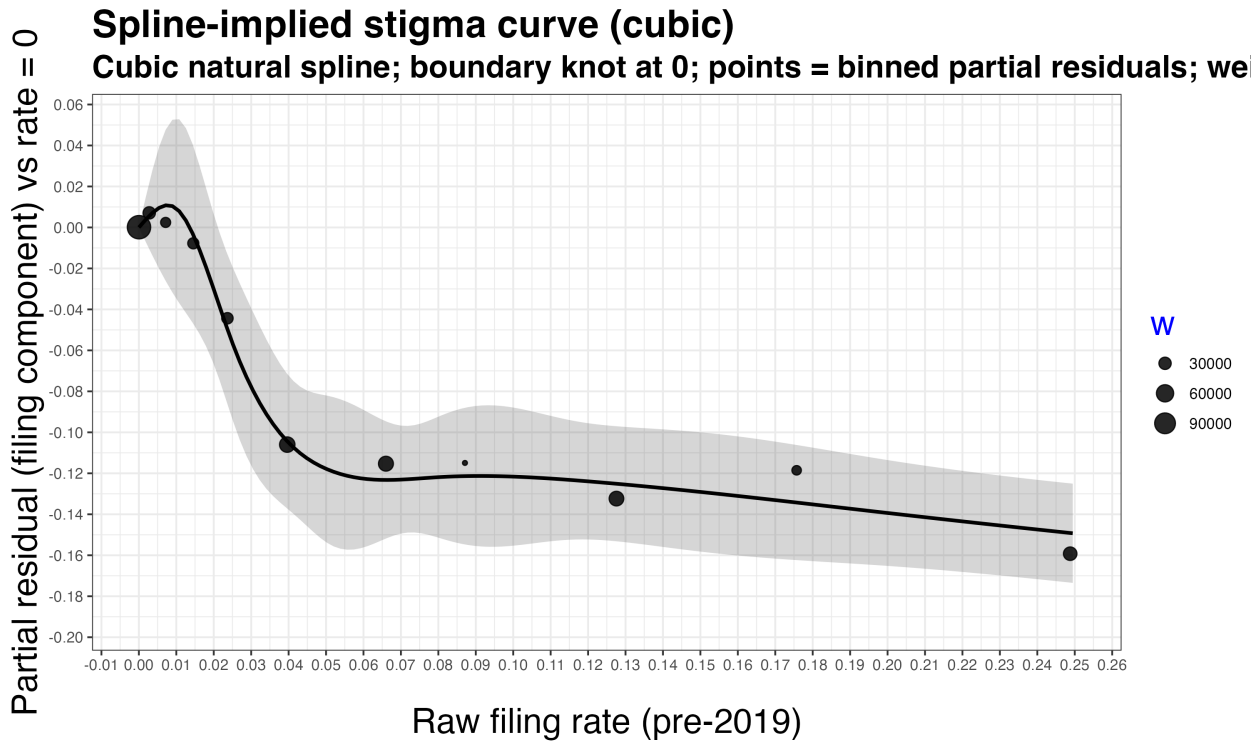

Figure 4: Spline-implied stigma curve with partial-residual binned means. The curve shows the fitted cubic natural spline component of log rent as a function of raw filing rate (relative to never-filers), with 95% confidence band. Points are binned means of partial residuals.

5.4 Spline Robustness: Linear vs Quadratic vs Cubic

To assess sensitivity of the stigma curve shape to the smoothness assumption, we estimate three spline specifications with identical knot placement (concentrated at low filing rates: 1%, 2.5%, 5%, 8%, 12%, 20%):

- **Linear (degree 1):** Piecewise-linear with kinks at each knot. Imposes no curvature within segments; slope changes occur only at knots.
- **Quadratic (degree 2):** Piecewise-quadratic; allows smooth curvature but constrains the second derivative to be piecewise-constant.
- **Cubic (degree 3, natural spline):** Piecewise-cubic with natural boundary constraints (linear beyond boundary knots). Most flexible within the interior.

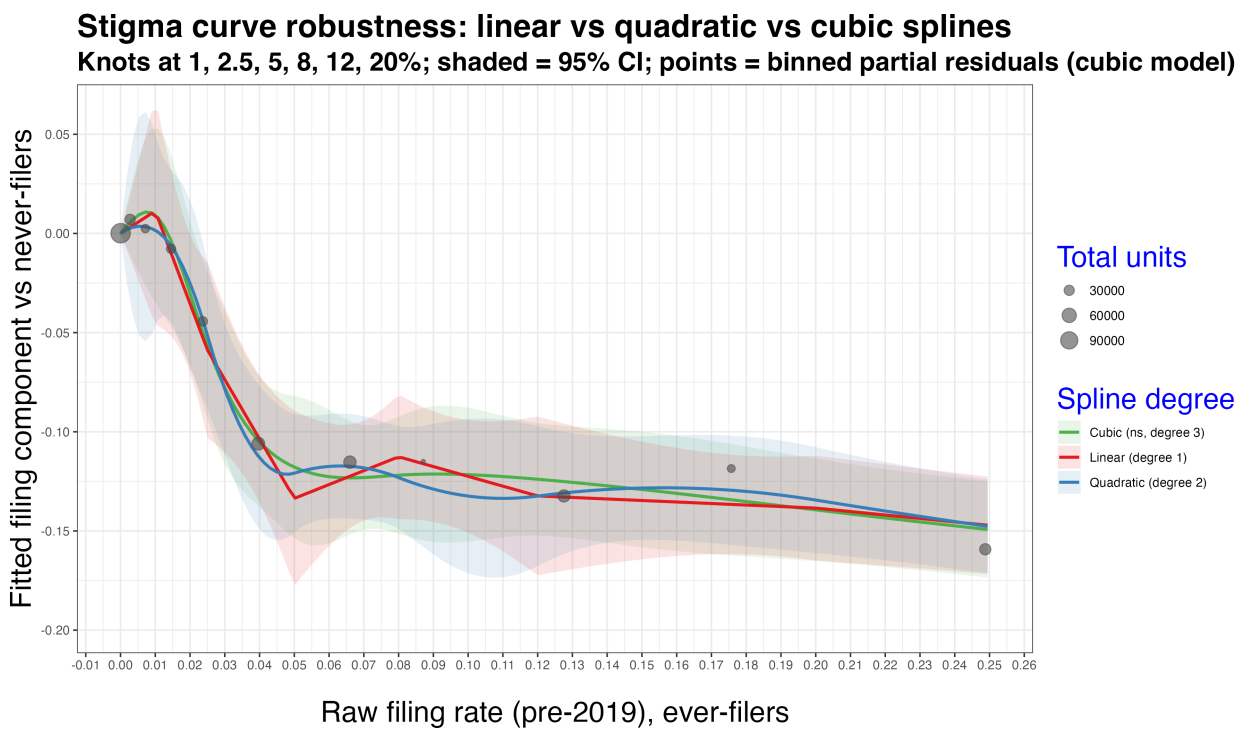

Figure 5: Comparison of stigma curves across spline degrees. All three specifications use the same knots concentrated at low filing rates. Shaded regions are 95% confidence bands (delta method). Points are binned partial residuals from the cubic model.

5.5 Hump Diagnostic: Never-Filer Composition

The spline curves show a hump near zero filing rate: buildings with very low positive rates appear to have *higher* rents than never-filers. This reflects composition differences — never-filers are 100% Altos-sourced (since eviction court records mechanically require a filing), include 25% single-family homes, and have a median of 1 unit. The barely-ever-filers ((0, 1%] bin) are large, high-quality apartment complexes (median 46 units, 26% quality-A).

To test whether the hump is a composition artifact, we re-estimate the cubic spline under progressively richer controls:

1. **Baseline** (no source FE — as in the main spline)
2. **+ Source FE** (absorbs Altos vs eviction-court rent level differences)
3. **+ Source FE + panel tenure** (controls for years observed pre-2019)
4. **Multi-family only** (drops SFH, which are 25% of never-filers)
5. **Multi-family + source + tenure** (all three adjustments)

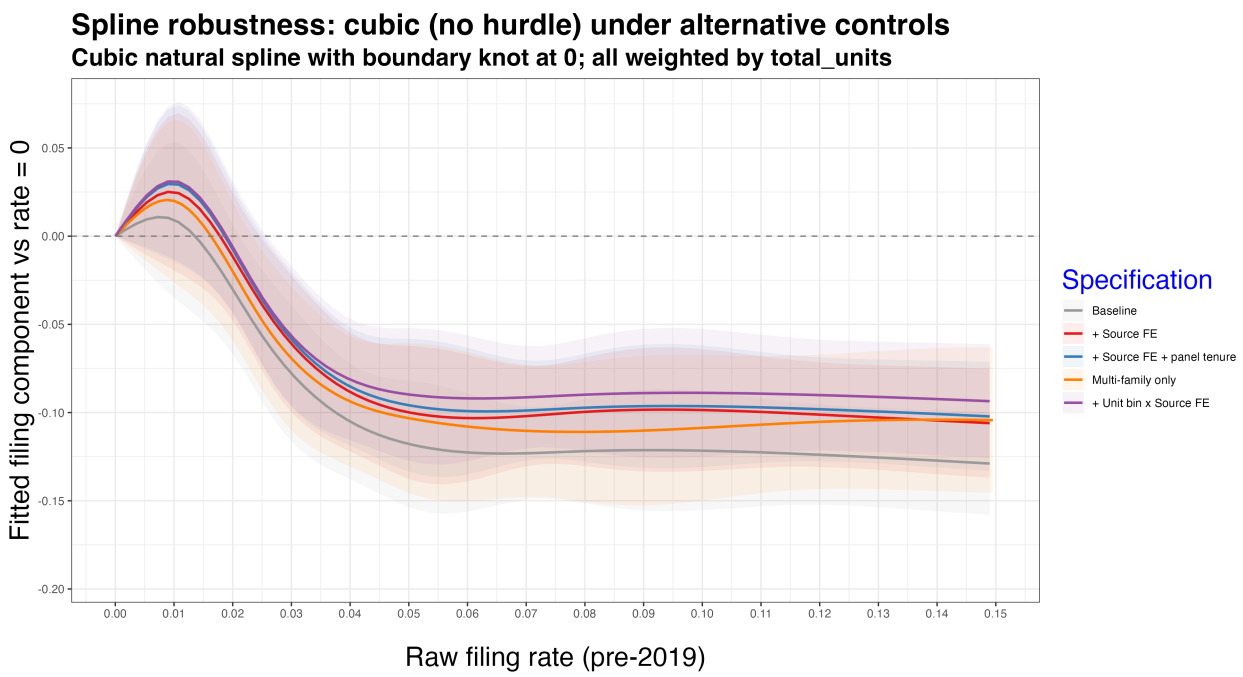

Figure 6: Hump diagnostic: cubic spline under alternative controls and sample restrictions. Hump attenuates with source FE and disappears when restricting to multi-family buildings.

6 Interpretation

The hedonic regressions estimate the **rent penalty** (or premium) associated with eviction filing intensity, conditional on neighborhood (tract FE) and building observables. Key quantities:

- **Binned model (Model 1):** The coefficient on the 20%+ bin relative to (5–10%] gives the rent differential for the highest-evicting buildings vs moderate-evicting buildings *within the same tract and building type*.
- **Continuous model (Model 2):** A one-unit increase in raw filing rate (e.g., from 0 to 1, which spans the entire range) is associated with a β-log-point change in rent. For typical variation (0 to 0.20), multiply by 0.20.
- **Tenant composition stability:** If adding tenant composition controls (Models 3–4) substantially attenuates the eviction intensity coefficients, it suggests the rent–eviction relationship partly reflects sorting by demographics. If coefficients are stable, the relationship is robust to observable tenant composition.
- **Complaint $\times$ high-eviction interaction (Model 5):** Tests whether buildings with both habitability complaints and high eviction rates have differentially lower rents, consistent with a "stigma" or quality channel.

6.1 Comparison with EB Estimates

The raw long-run filing rate differs from the Empirical Bayes version in `price-regs-eb.R`:

- **Raw rate:** Simple ratio of total filings to total units to years observed. Noisier for small buildings (few units, few years).
- **EB rate:** Shrinks extreme raw rates toward a prior (city- or ZIP-level mean). More stable for buildings with thin data.

Both should produce qualitatively similar results for large, well-observed buildings. Differences for small buildings reflect the EB shrinkage pulling extreme rates toward the mean.